This assignment will not be graded and is only for practice.

Multiple Choice Questions (MCQ):

1) Which of the following statements are correct? ***1 point***

- [ ] There exists a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(2,3) = (1,2)$, $T(1,-1) = (1,-1)$, and $T(4,1) = (1,0)$.
- [ ] If there is a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(2,3) = (1,2),$ and $T(1,-1) = (1,-1)$, then $T(x,y) = \frac{1}{5}(4x - y, -x + 4y)$.
- [ ] Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space V. Consider a linear transformation $T : V \to V$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. The matrix representation of T, with respect to β and γ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$
- [ ] Let $\{v_1, v_2, v_3\}$ be a basis of a vector space V and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space W. Consider a linear transformation $T : V \to W$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. Then, the rank of T is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If there is a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(2,3) = (1,2),$ and $T(1,-1) = (1,-1)$, then $T(x,y) = \frac{1}{5}(4x - y, -x + 4y)$.

Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space V. Consider a linear transformation $T : V \to V$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. The matrix representation of T, with respect to β and γ for the domain and codomain respectively, is $\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$

Let $\{v_1, v_2, v_3\}$ be a basis of a vector space V and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space W. Consider a linear transformation $T : V \to W$ such that $T(v_1) = 2v_3 + v_1, T(v_2) = 2v_1,$ and $T(v_3) = 2v_2$. Then, the rank of T is 3.

2) Consider the following set $S = \{(x,y) \mid x + y = 1, x, y \in \mathbb{R}\}$. In column A the matrix representation of some linear transformations are given with respect to the standard ordered bases. Match the entries in column A with the set $T(S)$ given in column B and the ***1 point***

geometric representations of S and $T(S)$ in column C.

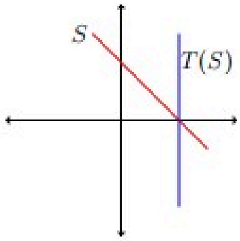
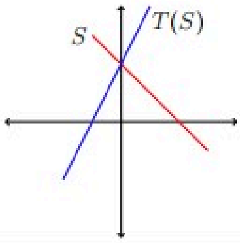
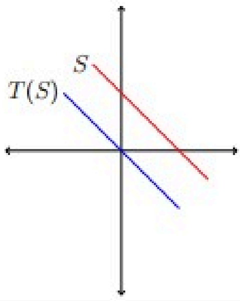
	Matrix form of linear transformation (Column A)		Image of the given set (Column B)		Geometric representations (Column C)
i)	$T:\mathbb{R}^2\to\mathbb{R}^2$ $\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$	a)	$T(S)=\{(x,y)\mid x=1, y\in\mathbb{R}\}$	1)	
ii)	$T:\mathbb{R}^2\to\mathbb{R}^2$ $\begin{bmatrix}0 & -1\\ 1 & -1\end{bmatrix}$	b)	$T(S)=\{(x,y)\mid x+y=0, x,y\in\mathbb{R}\}$	2)	
iii)	$T:\mathbb{R}^2\to\mathbb{R}^2$ $\begin{bmatrix}1 & 1\\ -1 & 0\end{bmatrix}$	c)	$T(S)=\{(x,y)\mid 2x-y=-1, x,y\in\mathbb{R}\}$	3)	

Table: M2W6P1

- [ ] i → b → 2.
- [ ] ii → c → 1.
- [ ] ii → c → 2.
- [ ] iii → a → 3.

- [ ] i → b → 3.
- [ ] iii → a → 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

ii → c → 2.
i → b → 3.
iii → a → 1.

Multiple Select Questions (MSQ):

3) Consider the following two groups. In Group 1, there are 3 functions from one vector space to another, and in Group 2, there are three properties of functions. All the vector spaces have usual addition and scalar multiplication. ***1 point***

Group-1

P: $T : \mathbb{R}^2 \to \mathbb{R}^2$ **such that** $T(x,y) = \begin{cases} (\frac{x^2}{y}, y) & \text{if } y = 0 \\ (x, 0) & \text{otherwise} \end{cases}$

Q: $T : \mathbb{R}^2 \to \mathbb{R}^2$ **such that** $T(x,y) = (x, xy)$

R: $T : \mathbb{R}^3 \to \mathbb{R}$ **such that** $T(x,y,z) = \frac{x+y+z}{5}$

Group-2

1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$
2: $T(cv) = cT(v)$ for all $v \in V$ and $c \in \mathbb{R}$.
3: T is a linear transformation

Choose the set of correct options.

- [ ] P satisfies 1 but not 2.
- [ ] P satisfies 2 but not 1
- [ ] Q satisfies 1
- [ ] Q satisfies 3
- [ ] P satisfies 3
- [ ] R satisfies 2
- [ ] R satisfies 3
- [ ] R satisfies 1 but not 2

No, the answer is incorrect.
Score: 0

Accepted Answers:

P satisfies 2 but not 1
R satisfies 2
R satisfies 3

Numerical Answer Type (NAT):

4) Suppose the matrix representation of a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with respect to the ordered bases $\beta = \{(1,0,0),(1,1,0),(1,1,1)\}$ for the domain and $\gamma = \{(1,1,1),(0,1,1),(0,0,1)\}$ for the range, is $I_{3\times3}$, i.e., the identity matrix of order 3. Let A be the matrix representation of the linear transformation T with respect to standard ordered basis of $\mathbb{R}^3$ for both domain and range. Then the determinant of A is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

5) Let T and S be two linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$, which are defined as follows:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \qquad S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x,y,z) = (x-y, y-z, z-x) \qquad S(x,y,z) = (x+y, y+z, z+x)$$

Let $S+T$ be defined to be the linear transformation as follows:

$$(S+T) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$(S+T)(x,y,z) = S(x,y,z) + T(x,y,z)$$

Let C be the matrix representation of $S+T$ with respect to the standard ordered bases of $\mathbb{R}^3$. If $C = nI$, where I denotes the identity matrix of order 3, then what will be the value of n?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

6) Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined as $T(x, y, z) = (x + 5y, z - 3x, y + 6z)$. Which of the following options are true about T? ***1 point***

- [ ] Nullity of T is 0.
- [ ] A basis for the range of T is $\{(1, 5, 0), (0, 1, 6)\}$.
- [ ] A basis for the range of T is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
- [ ] Rank of T is 2.
- [ ] The matrix representation of T with respect to the standard ordered bases of $\mathbb{R}^3$ is $\begin{bmatrix} 1 & -3 & 0 \\ 5 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.
- [ ] The matrix representation of T with respect to the standard ordered bases of $\mathbb{R}^3$ is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.
- [ ] Nullity of T is 1.
- [ ] Rank of T is 3.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Nullity of T is 0.
A basis for the range of T is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
The matrix representation of T with respect to the standard ordered bases of $\mathbb{R}^3$ is
$\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.
Rank of T is 3.

7) Let V be a vector space with ordered basis $\alpha = \{v_1, v_2, \ldots, v_{10}\}$. Consider a linear transformation $T : V \rightarrow V$ such that $T(v_1) = v_1$ and $T(v_i) = v_i - v_{i-1}$, where $i > 1$. Let A be the matrix representation of the linear transformation T, with respect to α for both the domain and codomain. what is the trace of A?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

Comprehension Type Question:

For each video on YouTube, consider the vector (Number of comments in thousands, Number of shares in thousands, Number of views in thousands) and consider the subset S of $\mathbb{R}^3$ consisting of these vectors. It is known that the ranking of a YouTube video can be thought of as the restriction of a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ to S where $\mathbb{R}^3$ comes with usual addition and scalar multiplication. (assume ranking 0 (zero) is possible and in practice negative rankings can be ignored).
The following data in Table M2W6P2 shows the correlation between comments, shares, and views - with YouTube rankings of a video.

Number of comments (in thousands)	Number of shares (in thousands)	Number of views (in thousands)	Ranking of a video
3	10	8	3
4	15	10	2
5	20	15	1

Table: M2W6P2

Use the above information to answer questions 8, 9 and 10.

8) Which of the following sets of vectors in S have the property that the YouTube video ***1 point*** corresponding to every linear combination of that set which is in S has ranking 0?

- [ ] $\{(1,1,0),(0,0,1)\}$
- [ ] $\{(6,25,0),(0,0,25)\}$
- [ ] $\{(25,6,0),(0,0,25)\}$
- [ ] $\{(12,50,0),(0,0,1)\}$
- [ ] $\{(24,100,0),(0,0,25)\}$
- [ ] $\{(6,0,0),(0,25,0),(0,0,1)\}$
- [ ] $\{(25,0,0),(0,6,0),(0,0,1)\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(6,25,0),(0,0,25)\}$
$\{(12,50,0),(0,0,1)\}$
$\{(24,100,0),(0,0,25)\}$

9) Which of the following sets of vectors in S have the property that the YouTube video ***1 point*** corresponding to every linear combination of that set which is in S has ranking a multiple of 15?

- [ ] $\{(3,0,0),(0,0,1)\}$
- [ ] $\{(3,0,1)\}$
- [ ] $\{(3,0,1),(0,0,1)\}$
- [ ] $\{(0,0,1)\}$
- [ ] $\{(9,25,0),(0,0,1)\}$
- [ ] $\{(25,9,0),(0,0,1)\}$
- [ ]

$\{(9,0,0),(0,25,0),(0,0,1)\}$

- [ ] $\{(25,0,0),(0,9,0),(0,0,1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(3,0,0),(0,0,1)\}$
$\{(3,0,1)\}$
$\{(3,0,1),(0,0,1)\}$
$\{(0,0,1)\}$
$\{(9,25,0),(0,0,1)\}$
$\{(9,0,0),(0,25,0),(0,0,1)\}$

10) Suppose that for a YouTube video there are 5 (in thousands) shares. What must be the number of comments (in thousands) so that the ranking of the video will be 4? [Note: Suppose your answer is 6000, then you have to enter 6 as the answer.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Your score is: 0/10